MILITARY SPECIFICATION

SIGNAL, SMOKE, GROUND
XM202 THRU XM205
RED, YELLOW, GREEN AND WHITE
PARTS FOR AND LOADING ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts and loading, assembling and packing of four (4) signals, smoke of different colors designated as Signal, Smoke, Ground, XM202 Red, XM203 Yellow, XM204 Green and XM205 White.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Packaging Methods of
MIL-P-223 - Powder, Black
MIL-P-20444 - Primer, Percussion, M42
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes (ABC-STD-105)

FSC: 1370

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MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-286 - Propellants, Solid: Sampling, Inspection and Testing
MIL-STD-331 - Fuze and Fuze Components; Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1234 - Pyrotechnics; Sampling, Inspection and Testing

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

8849726 - Primer, Percussion, M42G, Assembly
9280090 - Packing and Marking for Box, Wood Illumination Ground
9280092 - Packing and Marking for Shipping and Storage Container
9280095 - Packing and Marking for Sleeve
9293444 - Signal, Smoke, Ground, XM202 Thru XM205

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (dwg.) 9293444 and with all requirements specified in applicable specifications.

3.3 Moisture Content

3.3.1 Moisture Content of Smoke Composition. - The moisture content of each smoke composition at the loading station at the time of loading shall not exceed 0.1 percent.

3.3.2 Moisture Content of Black Powder. - The moisture content of the black powder at the loading station at the time of loading shall not exceed 0.3 percent.

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3.4 Functioning. - The signal assembly shall ignite and shall comply with the following requirements.

3.4.1 Premature. - The signal assembly shall not burst in the projector and shall not fail to reach a minimum height of fifteen (15) feet from the projector.

3.4.2 Burning Time. - The smoke assembly shall burn for not less than seven (7) seconds and shall burn out prior to hitting the ground.

3.4.3 Altitude. - The smoke assembly shall produce the correct color to a altitude of 400 ± 150 feet.

3.5 Water Immersion. - After the signals have been immersed in water, the signals shall comply with the functioning requirements.

3.6 Cold Temperature. - (Minus 65 plus or minus 5 degrees Fahrenheit (F.). - The signal assemblies after conditioning shall comply with the functioning requirements.

3.7 Hot Temperature. - (160 plus or minus 5 degrees Fahrenheit (F.). - The signal assemblies after conditioning shall comply with the functioning requirements.

3.8 Temperature and Humidity. - After the signals have been subjected to temperature and humidity conditioning, the signal assemblies shall comply with the functioning requirements.

3.9 Transportation Vibration. - The signals shall withstand transportation-vibration without exploding or burning. After the signals have been subjected to transportation-vibration, the signal assemblies shall comply with the functioning requirements.

3.10 Workmanship. - All parts shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of chips, burrs, sharp edges, cracks, dirt, grease, rust and other foreign materials. The metal parts shall not contain metal defects such as inclusions, cold shuts, or porous areas. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

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3.11 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification. In addition, a statement is required that the components and assemblies were dried and assembled as required by the applicable drawings.

4.2 Classification of Inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample (see 6.2b) as designated by the Contracting Officer. The first article sample shall consist of 25 complete sets of parts and sub-assemblies (i.e. 25 each of every component and sub-assembly and fifty (50) loaded assemblies of each color completely packaged and sealed as required by the applicable drawings shall be submitted to a Government proving ground disignated by the Contracting Officer. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3

4.3.2 Inspections to be performed. - See MIL-A-48078 and paras 4.4.2.1 through 4.4.2.15, testing per paras of 4.4.3 and Table 1 specified herein.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2 Cont.	Signal, Smoke, Ground XM202 thru XM205				9293444
					NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inert Parts				
	Examination for defects	25(a)		3.1 &	4.4.2.1 thru 4.4.2.15
	Signals (Complete Assemblies)				
	Examination for Defects	50(a)		3.2 &	4.4.2.9
	Tests				
	Water immersion - functioning	10(a)		3.5	4.5.2.1
	Cold Temperature - functioning	10(a)		3.6	4.5.2.2
	Hot Temperature - functioning	10(a)		3.7	4.5.2.3
	Temperature Humidity - functioning	10(a)		3.8	4.5.2.4
	Transportation-Vibration - Functioning	10(a)		3.9	4.5.2.5

NOTES:

(a) of each color assembly

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4.3.3 Rejection. - (See MIL-A-48078).

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Lot numbering as required shall be in accordance with MIL-STD-1168. In addition, inspection lots of signal assemblies shall contain:

a. Parts of one interfix lot number from one supplier.
b. Primers of one interfix lot number from one manufacturer.
c. Signals of one color only.
d. Composition ingredients from not more than one lot from one supplier.
e. Smoke composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.4.2 Examination. - See MIL-A-48078 (PA). - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cap				9248314
					NEXT HIGHER ASSEMBLY 9293444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A:					
101.	Crack or split		0.40%	3.2	Visual
Minor:					
201.	Outside diameter of case contact surface		0.65%	3.2	Gage
202.	Protective coating missing or incomplete		0.65%	3.2	Visual
203.	Overall depth		0.65%	3.2	Gage
204.	Evidence of poor workmanship (except cracks as defined above)		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9293446
4.4.2.2	Cup				NEXT HIGHER ASSEMBLY 9293444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Crack or split		100%	3.2	Visual
Major:	None defined				
Minor:					
201.	Total length		0.65%	3.2	Gage
202.	Thickness through bottom		0.65%	3.2	Gage
203	Outside diameter		0.65%	3.2	Gage
204.	Wall thickness		0.65%	3.2	Gage
205.	Evidence of poor workmanship (except cracks as defined above)		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Case, Signal, Smoke				9293448
					NEXT HIGHER ASSEMBLY 9293444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Crack or split		100%	3.2	Visual
2.	Color Incorrect		100%	3.2	Visual
Major A:					
101.	Flash hole obstructed		0.40%	3.2	Visual
102.	Pitch diameter of thread (max.)		0.40%	3.2	Gage
103.	Major diameter of thread (max)		0.40%	3.2	Gage
104.	Concentricity of primer cavity with thread		0.40%	3.2	Gage
Major B:					
131.	Diameter of charge cavity		0.40%	3.2	Gage
132.	Concentricity of inside diameter with outside diameter		0.40%	3.2	Gage
Minor:					
201.	Total length		0.65%	3.2	Gage
202.	Depth of primer cavity		0.65%	3.2	Gage
203.	Depth of charge cavity		0.65%	3.2	Gage
204.	Large outside diameter		0.65%	3.2	Gage
205.	Diameter of primer cavity		0.65%	3.2	Gage
206.	Outside diameter at primer end		0.65%	3.2	Gage
207.	Length to threaded end		0.65%	3.2	Gage
208.	Major diameter of thread, min		0.65%	3.2	Gage
209.	Pitch diameter of thread, min		0.65%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9293448
4.4.2.3	Case, Signal, Smoke				NEXT HIGHER ASSEMBLY 9293444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
210.	Bare spot in protective coating or coating missing		0.65%	3.2	Visual
211.	Evidence of poor workmanship (except cracks as defined above)		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9293447
4.4.2.4	Case Assembly				NEXT HIGHER ASSEMBLY 9293444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1.	Primer above flush		100%	3.2	Visual/Straight Edge
Major A: 101.	Sealing composition not 360° or where not required		0.40%	3.2	Visual
Minor: 201.	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Smoke Assembly (prior to loading applicable smoke composition)	1 OF 1	9293445
			NEXT HIGHER ASSEMBLY 9293444

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A:					
101.	Drop shot missing or inadequate		0.40%	3.2	Visual/Scale
102.	Presence of moisture or foreign material		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9293445
4.4.2.6	Smoke Assembly				NEXT HIGHER ASSEMBLY 9293444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Color identification missing		100%	3.2	Visual
2.	Color identification and smoke composition color do not coincide (color as applicable)		100%	3.2	Visual
Major A:					
101.	Any smoke composition increment missing (color as applicable)		0.40%	3.2	Visual
102.	Black powder missing		0.40%	3.2	Visual
103.	Assembly not drilled		0.40%	3.2	Visual
Minor:					
201.	Depth to charge, min.		0.65%	3.2	Gage
202.	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

MIL-S-48377 (PA)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Signal, Smoke, Ground (prior to assembling smoke assembly)	1 OF 1	9293444
			NEXT HIGHER ASSEMBLY N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A:					
101.	Propellant charge missing		0.40%	3.2	Visual
102.	Weight of charge		0.40%	3.2	Balance
Minor:					
201.	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48377 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Signal, Smoke, Ground (prior to assembling wad and cap)				9293444
					NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Color identification and case assembly color do not coincide		100%	3.2	Visual
2.	Smoke assembly binds in case assembly.		100%	3.2	Visual/Manaul
Major:	None defined				
Minor:					
201.	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9293444
4.4.2.9	Signal, Smoke, Ground				NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A:					
101.	Any component missing or improperly assembled		0.40%	3.2	Visual
Minor:					
201.	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Packing and Marking for box, set-up, inner prior to sealing				9280095
					NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B:					
131.	Tray damaged		0.40%	3.2	Visual
132.	Number of assemblies in tray incorrect		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48377 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.11	Packing and marking for box, set-up, inner, sealed		9280095
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Color identification missing or incorrect		100%	3.2	Visual
Major B:					
131.	Carton damaged to extent that contents are exposed or liable to become exposed		0.40%	3.2	Visual
Minor:					
201.	Contents loose		0.65%	3.2	Visual
202.	Marking misleading or unidentifiable (except as defined above)		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Packing and marking for box, set-up, outer, ammunition, packing for signal (prior to inserting into bag).				9280092
					NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Color identification missing or incorrect		100%	3.2	Visual
Major B:					
131.	Box damaged to extent that contents are exposed or liable to become exposed		0.40%	3.2	Visual
Minor:					
201.	Number of inner cartons incorrect		0.65%	3.2	Visual
202.	Sealing strip missing or insecure		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280092
4.4.2.13	Packing and marking for box, set-up, outer, ammunition, packing for signal (sealed)				NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Color identification missing or incorrect		100%	3.2	Visual
Major B:					
131.	Bag punctured, torn or seal incorrectly located		0.40%	3.2	Visual
Minor:					
201.	Marking misleading or unidentifiable (except as defined above)		0.65%	3.2	Visual
202.	Excess air in bag		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Packing and marking for box,wood, ammunition for signal (prior to closing)	1 OF 1	9280090
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	=None defined				
Major B:					
131.	Number of outer set-up boxes incorrect		0.40%	3.2	
132.	Any packing material missing		0.40%	3.2	
133.	Foreign material in box		0.40%	3.2	
Minor:					
201.	Evidence of poor workmanship		0.65%	3.10	

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Packing and marking for box, wood, ammunition for signal (sealed)				9280090
					NEXT HIGHER ASSEMBLY
					N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Color identification missing or incorrect		100%	3.2	Visual
Major B:					
131.	Box damaged exposing contents		0.40%	3.2	Visual
132.	Wire missing, broken or loose		0.40%	3.2	Visual/Manual
133.	Board broken or loose		0.40%	3.2	Visual/Manual
Minor:					
201.	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual
202.	Marking missing, misleading or unidentifiable (except as defined above)		0.65%	3.2	Visual
203.	Contents loose		0.65%	3.2	Visual
204.	Evidence of poor workmanship		0.65%	3.10	Visual

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4.4.3 Testing

4.4.3.1 Moisture Content (See Table II) -Major A Defects

TABLE II

Materials
Smoke composition (red, green, white and yellow) (see 3.3.1)
Black powder (see 3.3.2)
Packing material (see dwg. 9280092)

The contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each eight (8) hours production. A composite sample shall not be used. If the moisture content of the sample exceeds the requirements that quantity or sub-lot material represented by the sample shall not be used in production. If the quantity of material or sub-lot with excessive moisture has been used In loading and packing, the remaining unloaded and unpacked material shall not be used in production, and the loaded and packing signals shall be rejected. Test shall be performed as specified in 4.5.1.

4.4.3.2 Functioning (Proving Ground Test). - The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified In Table III, when tested as specified in 4.5.2 (destructive test).

TABLE III

Defects	Classification
Signal bursts in projector (see 3.4.1)	Critical
Altitude less than fifteen (15) feet (see 3.4.1)	Critical
Incorrect color	Critical
Signal fails to ignite (see 3.4)	Major A
Altitude less than two hundred and fifty (250) feet or more than 550 feet (see 3.4.3)	Major A
Burning time less than seven (7) seconds (see 3.4.2)	Major A
Signal Hits Ground Burning (see 3.4.2)	Major A

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4.4.3.2.1 Water Immersion (see 3.5)

4.4.3.2.1.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, one hundred and twenty-five (125) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if six (6) or more assemblies exhibit any of the defects listed in Table III.

4.4.3.2.1.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.4.3.2.1.1, fifty (50) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or of three (3) or more assemblies exhibit any of the defects listed in Table III.

4.4.3.2.2 Cold Temperature (see 3.6).

4.4.3.2.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirement specified, fifty (50) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if four (4) or more assemblies exhibit any of the defects listed in Table III.

4.4.3.2.2.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.4.3.2.2.1, thirty-two (32) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if three (3) or more assemblies exhibit any of the defects listed in Table III.

4.4.3.2.3 Hot Temperature (see 3.7).

4.4.3.2.3.1 Beginning with the first lot produced and continuing unitl three (3) consecutive lots have complied with the applicable requirement specified, fifty (50) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if four (4) or more assemblies exhibit any of the defects listed in Table III.

4.4.3.2.3.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.4.3.2.3.1, thirty-two (32) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if three (3) or more assemblies exhibit any of the defects listed in Table III.

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4.4.3.2.4 Temperature and Humidity (See 3.8). - Thirty-two (32) signal assemblies of each color shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if two (2) or more assemblies exhibit any of the defects listed in Table III, when tested as specified in 4.5.2.4 (destructive test).

4.4.3.2.5 Transportation-Vibration (see 3.9). - The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table IV, when tested as specified in 4.5.2.5 (Destructive test).

TABLE IV

Defect	Classification
Transporation-vibration safety failure (see 3.9)	Critical
Failure to meet function requirements after test (see 3.4)	See Table III

4.4.3.2.5.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, eighty (80) signal assemblies shall be selected from each lot for test. The lot shall be rejetted if a critical defect occurs or If four (4) or more assemblies exhibit any of the defects listed in Table III.

4.4.3.2.5.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.4.3.2.5.1, thirty-two (32) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if two (2) or more assemblies exhibit any of the defects listed in Table III.

4.4.3.3 Heat Seal of Packing Bag (see dwg. 9280092). - Major A Defect. - Ten (10) sealed packages shall be selected from each lot for test. If any heat seal fails to comply with the requirements, the lot shall be rejected. The test shall be as specified In 4.5.3 (non-destructive test).

4.4.3.4 Check-test for deterioration of primer, percussion, M42 (see MIL-P-20444). - If the total elapsed time between original acceptance of the M42 Primer lot and the assembly of that lot into the delay element exceeds two years or if the M42 Primers have been subjected to adverse conditions, however brief, at any

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time since previous tests, the applicable-lot shall be subjected to and must satisfactorily pass the check-test for deterioration specified in MIL-P-20444 except that 10 inches in lieu of 13 inches shall be utilized. The test shall be performed by the contractor on assemblies selected by the Government inspector, at the facility assembling the signals (see 6.5).

4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. (see Section 6 of MIL-A-48078 and section 6.3, herein.

4.5 Test Methods and Procedures

4.5.1 Moisture Content

4.5.1.1 Smoke Composition. - The method of determining the percentage of moisture in the smoke composition shall be in accordance with MIL-STD-1234, Method 102.1 except that the temperature shall be 130 degrees plus or minus 5 degrees Centigrade in lieu of 100 degrees Centigrade.

4.5.1.1 Alternate Method. - The moisture shall be determined in accordance with MIL-STD-286, Method 101.5 using Methyl Alcohol as the solvent.

4.5.1.2 Black Powder. - The method of determining the percentage of moisture in black powder shall be in accordance with MIL-P-223.

4.5.1.3 Packing Material. - The moisture content shall be determined in accordance with MIL-STD-1234, Method 102.1, except that a 10 gram sample shall be used and dried for two (2) hours.

4.5.2 Functioning. - The following functioning tests shall be performed at a Government Proving Ground.

4.5.2.1 Water immersion. - The signal assemblies shall be immersed in water at a temperature of 70 plus or minus 5 degrees F. to a depth of 10 to 12 inches for a minimum of 2 hours. Upon completion of water immersion, the signal assemblies shall be fired vertically from a fixture. The functioning shall be observed for compliance with the requirements.

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4,5.2.2 Cold Temperature (minus 65 plus or minus 5 degrees F.) - The signal assemblies shall be conditioned for a minimum of sixteen (16) hours at cold temperature and fired within 5 minutes after removal from the conditioning chamber by the same method specified in 4.5.2.1. The functioning shall be observed for compliance with the requirements.

4.5.2.3 Hot temperature (160 plus or minus 5 degrees F.). - The signal assemblies shall be conditioned for a minimum of sixteen (16) hours at hot temperature and fired within 5 minutes after removal from the conditioning chamber by the same method specified in 4.5.2.1. The functioning shall be observed for compliance with the requirements.

4.5.2.4 Temperature and Humidity. - The signal assemblies shall be subjected to a temperature and humidity cycle (see 6.7) as specified in MIL-STD-331, test 105. The signals shall be fired by the same method specified in 4.5.2.1. The functioning shall be observed for compliance with the requirements.

4.5.2.5 Transporation-Vibration. - The test shall be performed as specified in Test 104, Procedure II, of MIL-STD-331, at ambient temperature. The assembly, after test, shall be examined visually. If no degradation affecting safety or operability is apparent, the assembly shall be functioned as specified in 4.5.2.1 and observed for compliance with the requirements.

4.5.3 Heat Seal of Packing Bag. - The test shall be performed in accordance with the procedure specified in MIL-P-116. After test the seal shall be examined for compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A. - The signal assemblies shall be preserved and packaged in accordance with dwgs. 9280092 and 9280095.

5.2 Packing

5.2.1 Level A. - The packing carton shall be packed in accordance with dwg. 9280090.

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5.3 Marking. - Interior boxes, barrier bags and exterior wood boxes shall be marked in accordance with dwgs. 9280090, 9280092 and 9280095.

6. NOTES

6.1 Intended Use. - The signal is used with Projector MIL-P-48133.

6.2 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval - (see MIL-A-48078). Submit equipment designs, as required, to Commander: Picatinny Arsenals ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.4 Distribution of Ammunition Data Cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, Now Jersey 07801.

6.5 Check test for possible deterioration. - The following test is designed to prevent primers that have deteriorated from being assembled into major ammunition items. The test shall be normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Primer assemblies shall be selected from each lot for this test.

6.6 Proving Ground Functioning Test Summary

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Test	First Article Sample*	Production Lots		Requirements
		First three consecutive acceptable Sample*	Reduced Sampling*	
Water Immersion	10	125	50	3.5
Cold Temperature	10	50	32	3.6
Hot Temperature	10	50	32	3.7
Temp. & Humidity	10	32	32	3.8
Transp. Vibration	10	80	32	3.9

*Sample size designated is for each color signal.

6.7 Temperature and Humidity. - First article samples shall be subjected to a 14 day cycle. Production lots shall be subjected to a 3 day (72 hours) cycle.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1370-A-033

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POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-A
Dover, New Jersey 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBEF AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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